MULTI-CLOUD FDE STACK #07

MULTI-CLOUD FDE STACK · ISSUE 07 OF 08

See it before the customer does.

OpenTelemetry, Prometheus, Grafana and structured logs.

07

AR

Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

One telemetry path

1. App + OpenTelemetry SDK
2. OTLP to Collector
3. Redact + sample
4. Prometheus / traces / logs
5. Grafana + alerts

> The collector keeps backends swappable, and redacts personal data before it leaves the network.

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Akhilesh Ranjan Singh
github.com/iarsingh

Signals for an AI service

01 RED for the API

Request rate, error rate, p95 latency per route.

02 LLM calls

Latency, errors, tokens and cost by provider and model.

03 Quality

Retrieval hit rate, guardrail blocks, human overrides.

04 Dependencies

Database pool, queue depth, oldest message age.

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Akhilesh Ranjan Singh
github.com/iarsingh

Two queries worth knowing

PROMQL

```
# error ratio
sum(rate(http_requests_total{status=~"5.."}[5m]))
  / sum(rate(http_requests_total[5m]))

# p95 latency
histogram_quantile(0.95,
  sum by (le) (
    rate(http_request_duration_seconds_bucket[5m])))
```

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github.com/iarsingh

Telemetry habits

DO

- ✓ JSON logs with trace_id
- ✓ Labels with few values
- ✓ Alert on SLO burn rate
- ✓ Runbook link on every alert

AVOID

- ✗ user_id as a metric label
- ✗ Raw prompts in spans
- ✗ Alert on every error
- ✗ Dashboards nobody owns

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Before go-live

- [x] Traces across every hop
- [x] SLOs agreed with the customer
- [x] Burn-rate alerts routed
- [x] PII redacted in the collector
- [x] Runbooks tested in a game day

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KEY TAKEAWAY

If you cannot see it, you cannot support it.

→ One OpenTelemetry pipeline, any backend

→ Watch AI quality, not just uptime

→ Redact before data leaves the network

YOUR TURN

What is the one metric you add to every AI service?

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Next in the series: #08 Ship to any cloud from one pipeline

Save · Repost

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Akhilesh Ranjan Singh
github.com/iarsingh